

CSA16 – DATA WAREHOUSING AND DATA MINING - REAL-WORLD PROBLEM-BASED ASSIGNMENT

Intelligent Consumer-Trust Analytics for India's E-Commerce & Quick-Commerce Ecosystem

Course Outcomes	CO1: Design an appropriate dimensional data warehouse for decision support. CO2: Apply data cleaning, integration, transformation, reduction and discretization to heterogeneous real-world data. CO3: Evaluate classification models for reliable, explainable and actionable prediction.
Bloom's Taxonomy	L3 – Apply L4 – Analyze L5 – Evaluate L6 – Create
SDG Mapping	SDG 8 – Decent Work and Economic Growth SDG 9 – Industry, Innovation and Infrastructure SDG 12 – Responsible Consumption and Production

REAL-WORLD PROBLEM STATEMENT / CHALLENGE

India's rapid growth of e-commerce and quick-commerce has created a difficult data-driven consumer-protection problem. In August 2026, a Lok Sabha response reported that the National Consumer Helpline received 5,11,198 e-commerce grievances in 2025, compared with 2,24,592 in 2021; reported grievance categories include non-refund, defective/damaged/wrong products, non-delivery, missing products, delayed delivery and fake/spurious products. The same parliamentary response specifically noted complaints concerning quick-commerce platforms and highlighted the need for stronger controls around product authenticity, delivery, returns, refunds and grievance redressal. Therefore, assume that a large Indian omnichannel retailer/quick-commerce network has millions of daily records distributed across order management, seller, inventory, warehouse/dark-store, delivery, payment, return/refund and customer-support systems. Management currently reacts to complaints after consumers report problems. The central research question is: How can an integrated Data Warehouse and Data Mining framework predict high-risk orders, sellers, products, delivery zones and complaint patterns BEFORE a consumer grievance escalates, while avoiding unfairly flagging genuine customers?

Required Innovation – Students must go beyond routine sales analysis.

1. Construct a Star/Snowflake warehouse integrating at least six heterogeneous sources: Orders, Customers, Products/Sellers, Inventory, Delivery, Payments and Returns/Complaints.
2. Design an ETL pipeline and document data-quality problems such as duplicates, missing values, inconsistent seller/product identifiers, abnormal timestamps and conflicting refund statuses.
3. Create OLAP cubes and demonstrate Roll-up, Drill-down, Slice, Dice and Pivot to identify geographic complaint hotspots, seller/product risk, delivery failures and refund bottlenecks.
4. Engineer an original Consumer Trust Risk Index using measurable features such as complaint-to-order ratio, refund-delay rate, return frequency, delivery deviation, seller defect rate, product-category risk, order velocity and repeated complaint patterns.
5. Build and compare at least THREE classifiers (e.g., Decision Tree, Naïve Bayes, SVM) to predict a defined target such as High-Risk Order / Likely Complaint / Seller Risk. Accuracy alone is NOT sufficient.
6. Evaluate using Confusion Matrix, Precision, Recall, F1-score, Cross-validation and, where supported, ROC-AUC. Explain the cost of false positives versus false negatives in a consumer-protection setting.
7. Develop an actionable early-warning dashboard in KNIME and/or Weka showing which orders/sellers/products/locations require investigation and WHY.
8. Conduct an ethics and robustness analysis: privacy, explainability, class imbalance, potential bias against locations/customer groups, data leakage and the danger of automatically denying legitimate refunds.
9. Propose a novel intervention strategy: for example, risk-based quality inspection, seller audit prioritization, proactive refund verification, delivery-zone monitoring or complaint-prevention alerts. Quantify how your intervention could reduce unresolved complaints or operational loss.

EXPECTATION

Do not merely implement algorithms. State assumptions, formulate a defensible analytical hypothesis, justify the warehouse grain and dimensions, compare competing models experimentally, interpret unexpected results, and demonstrate how the proposed system would change an actual managerial decision. The final submission must contain architecture/schema, ETL workflow, OLAP evidence, feature-engineering rationale, model comparison, error analysis, dashboard/visualizations and a short evidence-based recommendation.

Contemporary Indian context: Lok Sabha / Ministry of Consumer Affairs data reported in August 2026 on e-commerce and quick-commerce grievances; National Consumer Helpline reported ₹52 crore in refunds across 31 sectors during Apr 2025–Jan 2026.

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CSA1603 – DATA WAREHOUSING AND DATA MINING

UNIQUE ASSIGNMENT

Intelligent Consumer-Trust Analytics for India's E-Commerce & Quick-Commerce Ecosystem

1. INTRODUCTION

The rapid growth of e-commerce and quick-commerce platforms has increased the importance of customer trust, delivery reliability, product quality and effective complaint management. Customers expect accurate orders, timely deliveries, satisfactory service and quick resolution of refund-related issues.

However, large volumes of transactional data make it difficult for organizations to manually identify customers, products, platforms and orders associated with higher consumer-trust risks.

This project proposes an **Intelligent Consumer-Trust Analytics framework** that combines data warehousing, ETL, OLAP analysis, feature engineering and machine-learning techniques to identify and analyse potentially high-risk orders.

A **Consumer Trust Risk Index (CTRI)** is developed using measurable indicators such as complaint behaviour, refund behaviour, delivery deviation, product-category risk, platform risk and repeated customer issues.

The project also compares three classification algorithms:

- Decision Tree
- Naïve Bayes
- Support Vector Machine (SVM)

The resulting analytical framework can support organizations in monitoring consumer-trust issues and prioritizing orders requiring further investigation.

2. PROBLEM STATEMENT

E-commerce and quick-commerce organizations generate large amounts of customer, order, delivery, product, platform and complaint-related information. However, this information is often difficult to analyse collectively.

The main problem addressed in this project is:

How can heterogeneous e-commerce transaction data be transformed into a data warehouse and analytical framework capable of identifying high consumer-trust-risk orders and supporting reliable managerial decisions?

The project focuses on integrating available customer and order-level information, performing ETL and preprocessing, developing a Consumer Trust Risk Index, applying OLAP operations and comparing machine-learning classification models.

3. OBJECTIVES

The major objectives are:

1. To design a **star-schema data warehouse** for consumer-trust analytics.
2. To perform data cleaning, transformation and feature engineering.
3. To analyse customer, platform, product and delivery behaviour.
4. To construct a **Consumer Trust Risk Index (CTRI)**.
5. To classify orders into Low, Medium and High risk categories.
6. To perform OLAP operations such as:
 - Roll-up
 - Drill-down
 - Slice
 - Dice
 - Pivot
7. To compare Decision Tree, Naïve Bayes and SVM classifiers.
8. To evaluate models using:
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - ROC-AUC
9. To perform 5-fold stratified cross-validation.
10. To identify data-quality, leakage, privacy and ethical issues.
11. To provide actionable recommendations for improving consumer trust.

4. DATASET DESCRIPTION

The supplied dataset contains **100,000 records and 11 attributes**.

Dataset attributes

Attribute	Description
Order ID	Unique order identifier
Customer ID	Unique customer identifier
Platform	E-commerce/quick-commerce platform
Order Date & Time	Order timing information
Delivery Time (Minutes)	Delivery duration
Product Category	Product category
Order Value (INR)	Monetary value of order
Customer Feedback	Customer feedback text
Service Rating	Customer service rating
Delivery Delay	Indicates whether delivery was delayed
Refund Requested	Indicates whether a refund was requested

Dataset size

Rows:100,000

Columns: 11

Data quality

The initial analysis showed:

- Missing values: **0**
- Duplicate rows: **0**
- Duplicate Order IDs: **0**

After cleaning:

- Rows before cleaning: **100,000**
- Rows after cleaning: **100,000**
- Duplicate rows after cleaning: **0**
- Duplicate Order IDs: **0**

Therefore, the dataset did not require removal of records because of missing or duplicate observations.

5. ASSUMPTIONS AND ANALYTICAL HYPOTHESIS

5.1 Assumptions

The following assumptions were made:

1. Each Order ID represents one unique transaction.
2. Customer ID identifies the customer associated with an order.
3. Delivery Delay indicates whether the order experienced a delivery delay.
4. Refund Requested is treated as an indicator of refund-related consumer dissatisfaction.
5. Service Rating represents customer satisfaction.
6. Platform is used as the available organizational/platform dimension because the dataset does not contain a separate Seller ID.
7. Refund-delay rate is treated as a proxy because actual refund-processing timestamps are unavailable.
8. Return frequency is approximated using refund behaviour because a separate return field is unavailable.

5.2 Hypothesis

Orders exhibiting combinations of delivery delays, refund behaviour, negative feedback, low service ratings, repeated customer issues, category risk and platform-level risk are more likely to belong to a High Consumer Trust Risk category.

6. SYSTEM ARCHITECTURE

The proposed architecture consists of the following stages:

Source Dataset → ETL → Data Cleaning → Feature Engineering → Data Warehouse → OLAP Analysis → CTRI Calculation → Machine Learning → Dashboard & Decision Support

Architecture components

1. Data source
2. Data preprocessing
3. Feature engineering
4. Star-schema warehouse
5. OLAP analysis
6. CTRI computation
7. Machine-learning models
8. Model evaluation
9. Dashboard

10. Managerial intervention

7. ETL AND DATA PREPROCESSING

7.1 Extract

The dataset was loaded into Python using Pandas.

The dataset contains:

100,000 rows × 11 columns

7.2 Transform

The following transformations were performed:

Data cleaning

- Duplicate rows were checked.
- Duplicate Order IDs were checked.
- Numeric columns were converted to numeric format.
- Missing numeric values were handled using median values.
- Missing categorical values were replaced with "Unknown".
- Text fields were stripped of unnecessary spaces.
- Yes/No fields were standardized.

Date/time processing

The Order Date & Time field was converted into a datetime representation where possible.

Additional temporal attributes were derived, including:

- Hour
- Minute
- Day
- Month
- Weekday

7.3 Feature Engineering

The following customer-level features were created:

- Customer Order Count
- Customer Total Value
- Customer Average Order Value

- Customer Refund Count
- Customer Delay Count
- Customer Refund Rate
- Customer Delay Rate

Additional features included:

- Delivery Delay Flag
- Refund Flag
- Product Category Risk
- Platform Risk
- Delivery Deviation
- Refund Delay Rate
- Return Frequency
- Order Velocity
- Complaint Pattern
- Complaint Flag

8. CONSUMER TRUST RISK INDEX (CTRI)

A major innovation of the project is the development of the **Consumer Trust Risk Index (CTRI)**.

The index combines eight measurable components.

CTRI formula

$$CTRI = 0.20(CRS) + 0.15(RDRS) + 0.15(RFRS) + 0.15(DDS) + 0.15(SDS) + 0.10(PRS) + 0.05(OVS) + 0.05(RCS)$$

where:

Component	Weight
Complaint Ratio Score	20%
Refund Delay Score	15%
Return Frequency Score	15%
Delivery Deviation Score	15%

Seller/Platform Defect Score	15%
Product Risk Score	10%
Order Velocity Score	5%
Repeated Complaint Score	5%

The resulting score is scaled to a **0–100** range.

9. RISK CLASSIFICATION

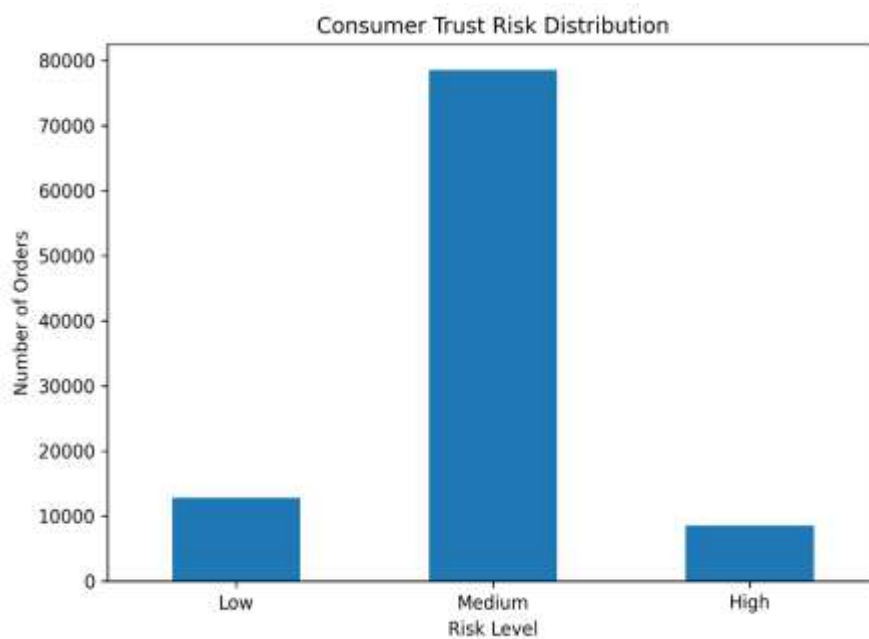
The CTRI score is converted into three risk levels:

CTRI Score	Risk Level
≤ 30	Low
31–60	Medium
> 60	High

Actual risk distribution

Risk Level	Orders	Percentage
Low	12,829	12.83%
Medium	78,599	78.60%
High	8,572	8.57%
Total	100,000	100%

Therefore, **8,572 orders** were classified as **High Risk**.



10. DATA WAREHOUSE DESIGN

A **star schema** was implemented.

Fact Table

Fact_Order_Trust

The fact table contains **100,000 records and 16 attributes**.

It stores transaction-level measures and foreign keys connecting the dimension tables.

Dimension Tables

Dim_Customer

Contains customer-level analytical information.

Records: 9,000

Important attributes include:

- Customer Key
- Customer ID
- Customer Order Count
- Customer Total Value
- Customer Average Value
- Customer Refund Rate
- Customer Delay Rate

Dim_Product

Contains product-category information.

Records: 6

Dim_Platform

Contains platform-level information.

Records: 3

The three platforms represented are:

- Blinkit
- JioMart
- Swiggy Instamart

Because the dataset does not contain Seller ID, **Dim_Platform is used instead of Dim_Seller**.

Dim_Delivery

Contains delivery-related information.

Records: 69

Dim_Date

Contains temporal attributes.

Records: 25

Dim_Complaint

Contains complaint-related categories/patterns.

Records: 10

11. WAREHOUSE INTEGRITY VALIDATION

Foreign-key validation was performed after warehouse creation.

Foreign Key	Invalid Rows
Customer Key	0
Product Key	0
Platform Key	0
Delivery Key	0
Date Key	0
Complaint Key	0

This demonstrates that all fact-table foreign-key mappings were successfully established.

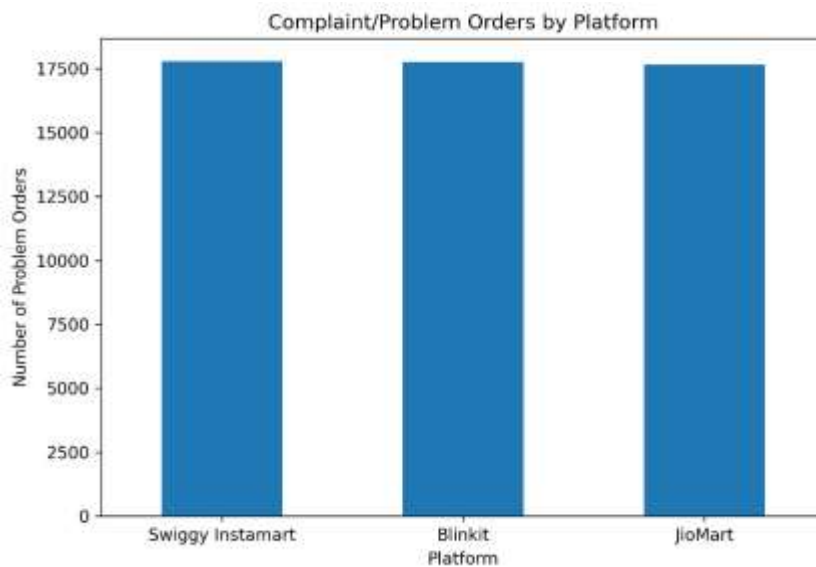
12. OLAP ANALYSIS

The project implements five major OLAP operations.

12.1 OLAP Roll-up

Roll-up was performed from:

Platform → Product Category



Results:

Platform	Category	Orders	Total Value	Refunds	High Risk
Blinkit	Beverages	5,593	₹3,063,913	2,547	46
Blinkit	Dairy	5,643	₹2,547,605	2,606	41
JioMart	Beverages	5,411	₹2,967,101	2,496	949
JioMart	Grocery	5,605	₹4,746,467	2,623	1,028
Swiggy Instamart	Grocery	5,546	₹4,716,086	2,591	439

Interpretation

JioMart shows considerably higher high-risk order counts across the displayed categories. For example, JioMart Grocery contains **1,028 high-risk orders**, compared with 38 for Blinkit Grocery and 439 for Swiggy Instamart Grocery.

This indicates that platform-level risk is an important dimension for further investigation.

13. OLAP DRILL-DOWN

Drill-down was performed from:

Platform → Product Category → Risk Level

For example, Blinkit results include:

Category	High	Low	Medium
Beverages	46	1,914	3,633
Dairy	41	1,905	3,697
Fruits & Vegetables	37	1,821	3,718

Grocery	38	1,812	3,736
Personal Care	57	1,859	3,547
Snacks	45	1,935	3,583

Interpretation

Blinkit Personal Care has the highest high-risk count among the displayed Blinkit categories, with **57 high-risk orders**.

14. OLAP SLICE

A slice operation was performed by selecting:

Platform = Blinkit

The resulting dataset contains only Blinkit orders.

Example:

Order ID	Product Category	Order Value	Risk
ORD000002	Dairy	₹279	Low
ORD000005	Beverages	₹334	Medium
ORD000006	Personal Care	₹1,939	Medium
ORD000013	Fruits & Vegetables	₹516	Low

The slice allows management to focus on a single platform.

15. OLAP DICE

A dice operation was performed using multiple conditions:

- Platform = Blinkit
- Risk Level = High
- Refund Requested = Yes

The resulting records represent orders satisfying all three conditions.

Examples include:

Order ID	Category	Order Value	CTRI
ORD000235	Snacks	₹391	64.35
ORD000294	Dairy	₹564	60.76
ORD000600	Snacks	₹383	60.21
ORD001614	Personal Care	₹1,881	62.74

This helps management identify specific orders that may require closer investigation.

16. OLAP PIVOT

The pivot operation analysed:

Platform × Product Category

Platform	Beverages	Dairy	Fruits & Vegetables	Grocery	Personal Care	Snacks
Blinkit	5,593	5,643	5,576	5,586	5,463	5,563
JioMart	5,411	5,586	5,546	5,605	5,436	5,543
Swiggy Instamart	5,532	5,628	5,510	5,546	5,634	5,599

The pivot provides a compact comparison of order volumes across platforms and product categories.

17. MACHINE LEARNING

Three classification models were implemented:

1. Decision Tree
2. Naïve Bayes
3. Support Vector Machine

The target variable was:

High_Risk_Order

where an order is considered high risk when:

$$CTRI \geq 60$$

18. TRAINING AND TESTING

The dataset was divided using stratified sampling:

- Training data: **80,000**
- Testing data: **20,000**

Stratification was used to maintain the class distribution between training and testing sets.

19. MODEL EVALUATION

Decision Tree

- Accuracy: **99.24%**
- Precision: **92.49%**

- Recall: **99.18%**
- F1-score: **95.72%**
- ROC-AUC: **99.69%**

Naïve Bayes

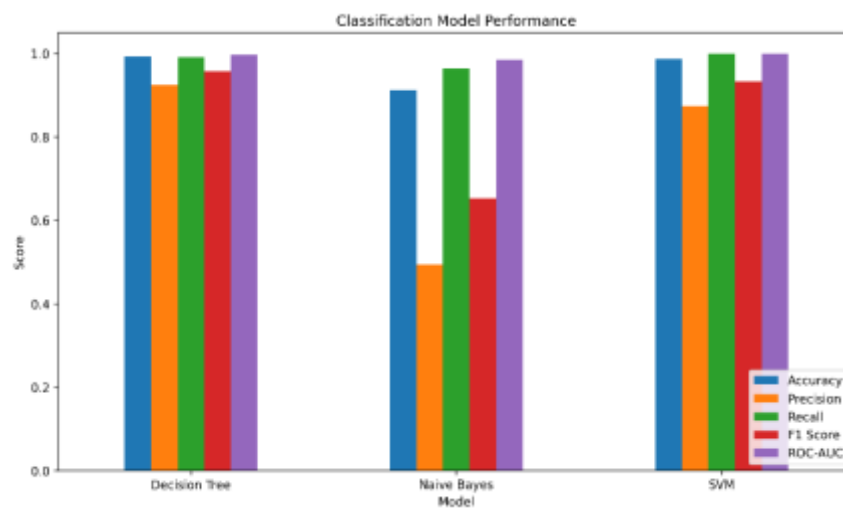
- Accuracy: **91.22%**
- Precision: **49.36%**
- Recall: **96.44%**
- F1-score: **65.30%**
- ROC-AUC: **98.45%**

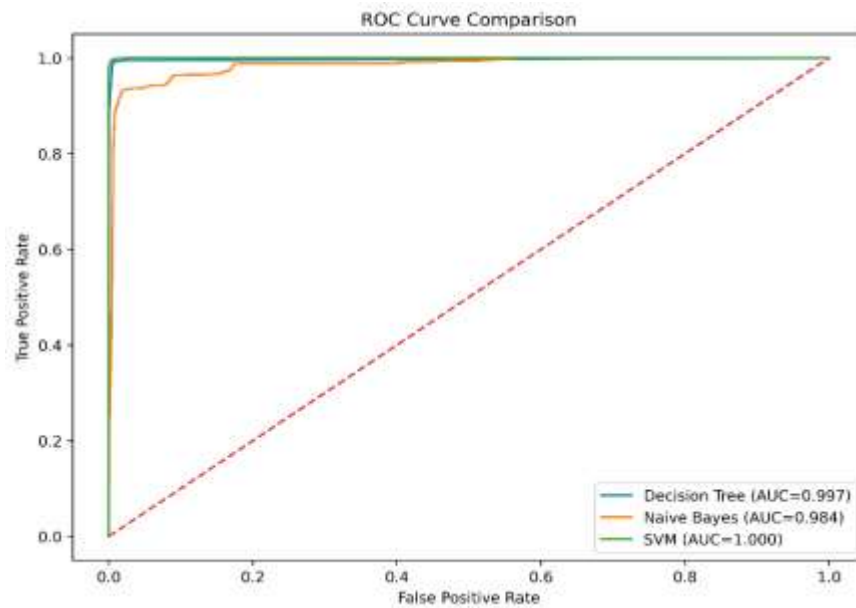
SVM

- Accuracy: **98.76%**
- Precision: **87.40%**
- Recall: **99.94%**
- F1-score: **93.25%**
- ROC-AUC: **99.99%**

20. MODEL COMPARISON

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Decision Tree	99.24%	92.49%	99.18%	95.72%	99.69%
SVM	98.76%	87.40%	99.94%	93.25%	99.99%
Naïve Bayes	91.22%	49.36%	96.44%	65.30%	98.45%





Best Model

The **Decision Tree** is selected as the best overall model because it achieved:

- Highest accuracy
- Highest precision
- Highest F1-score
- Strong recall
- Strong ROC-AUC
- Better interpretability compared with SVM

Therefore:

Decision Tree is selected as the preferred analytical benchmark model for this project.

21. CROSS-VALIDATION

Five-fold stratified cross-validation was performed.

Model	Mean F1	Std. F1
Decision Tree	0.9590	0.0022
SVM	0.9338	0.0023
Naïve Bayes	0.6482	0.0066

The Decision Tree again achieved the highest average F1-score.

The relatively small standard deviation indicates stable performance across the five folds.

22. CONFUSION MATRIX ANALYSIS

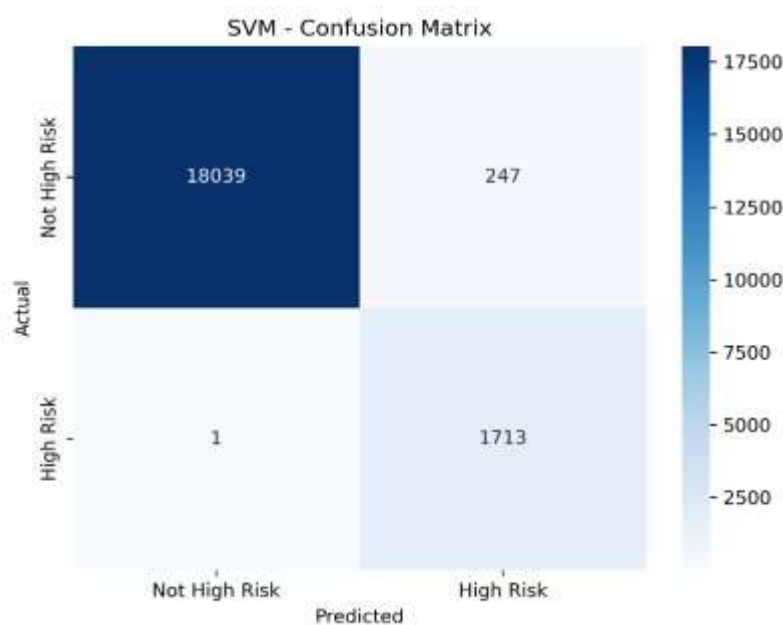
The confusion matrices were generated for all three models.

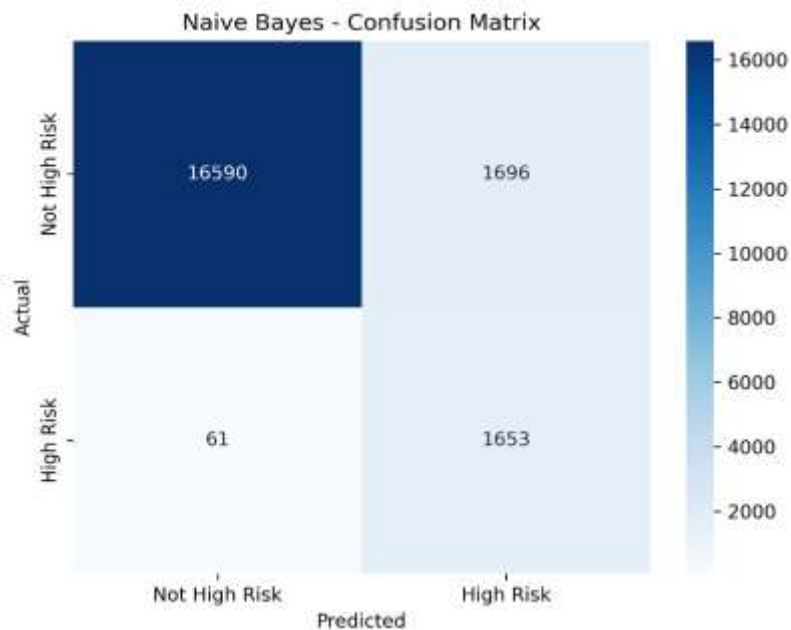
The classification report for the Decision Tree shows:

Class	Precision	Recall	F1
Not High Risk	1.00	0.99	1.00
High Risk	0.92	0.99	0.96

The model has a very high recall for High Risk orders.

This is important because a **False Negative** means a genuinely high-risk order is not identified.





23. FALSE POSITIVE VS FALSE NEGATIVE

In consumer-trust analytics, the two errors have different consequences.

False Positive

A normal order is incorrectly classified as high risk.

Possible consequence:

- Unnecessary investigation
- Additional operational cost
- Customer inconvenience

False Negative

A genuinely high-risk order is classified as normal.

Possible consequence:

- Consumer dissatisfaction may remain unresolved.
- Repeated complaints may occur.
- Trust may decrease.
- The organization may lose customer loyalty.

Therefore, recall is particularly important when the objective is early identification of potential consumer-trust problems.

24. DATA LEAKAGE AND MODEL VALIDATION

An important limitation was identified during the implementation.

The target:

High_Risk_Order

is derived from:

CTRI

Several variables used by the machine-learning models are also components of the CTRI calculation.

Therefore, the very high model performance may be partly influenced by **target leakage**.

The current results should therefore be treated as an:

Analytical benchmark rather than a leakage-free production prediction model.

For production deployment:

- Future complaint outcomes should not be used.
- Future refund decisions should not be used.
- Post-delivery feedback should not be used if prediction occurs before delivery.
- Only information available at the prediction point should be included.

This is an important methodological limitation and should be explicitly reported.

25. DASHBOARD AND DECISION SUPPORT

The project generates a dashboard-ready dataset containing:

- Order ID
- Customer ID
- Platform
- Product Category
- Order Value
- Delivery Time
- Service Rating
- Delivery Delay
- Refund
- CTRI
- Risk Level
- High Risk Flag

- Complaint Flag

The dashboard can be used to monitor:

1. High-risk orders
2. Platform risk
3. Product-category risk
4. Refund patterns
5. Delivery delays
6. Complaint patterns
7. Customer-level behaviour

26. HIGH-RISK ORDER ANALYSIS

The analysis identified **8,572 high-risk orders**.

Examples of high-risk records include orders with combinations such as:

- Delivery delay
- Refund requested
- Low service rating
- High CTRI

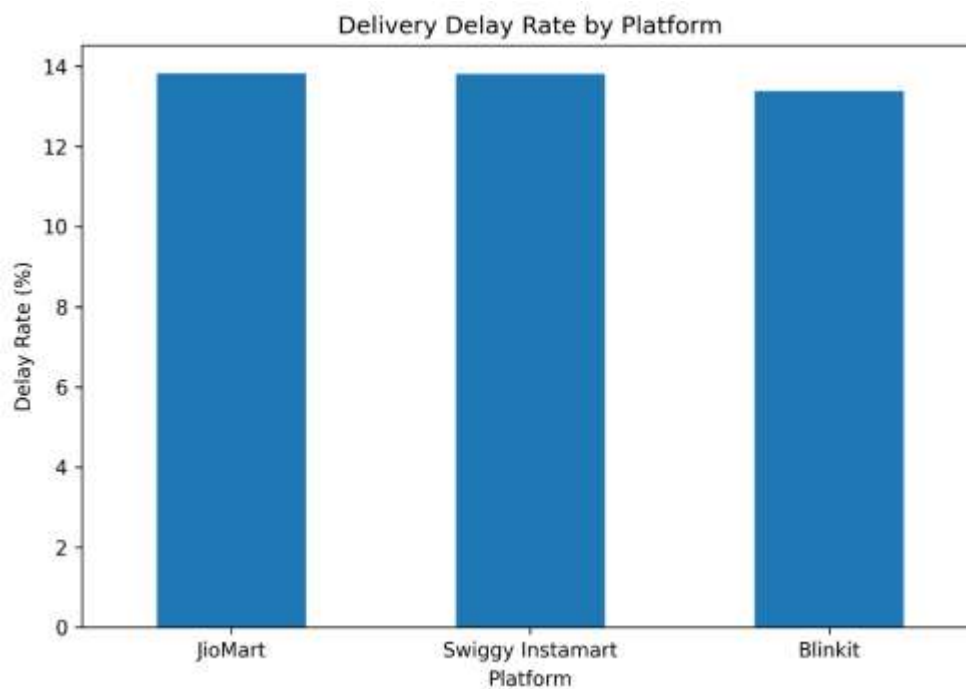
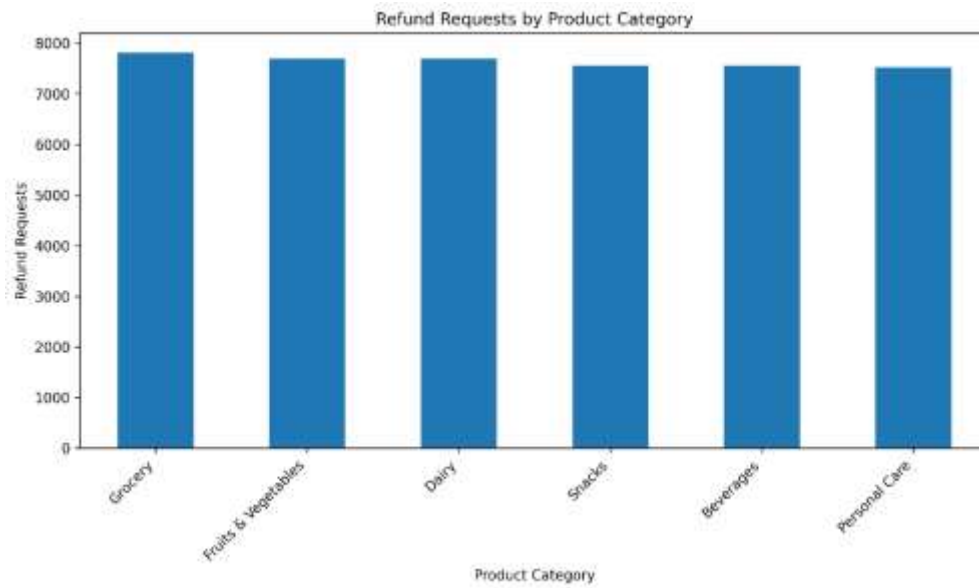
For example:

ORD008801

was identified as:

- Platform: JioMart
- Category: Beverages
- Risk Level: High
- Delivery Delay: Yes
- Refund Requested: Yes
- Service Rating: 1

Such records can be prioritized for operational review.



27. PROPOSED INTERVENTION STRATEGY

A three-level intervention strategy is proposed.

Low Risk

Action: Normal processing

- No additional intervention.
- Continue standard monitoring.

Medium Risk

Action: Proactive monitoring

- Monitor delivery progress.
- Check repeated customer issues.
- Prioritize customer-service alerts.

High Risk

Action: Early investigation

- Prioritize customer-service review.
- Investigate delivery problems.
- Review refund/complaint history.
- Trigger proactive communication where appropriate.

Importantly:

The model should support human decision-making and should **not automatically deny refunds or penalize customers**.

28. ETHICAL CONSIDERATIONS

Privacy

Customer IDs and behavioural information should be protected.

Explainability

Decision Tree predictions are relatively easier to explain to business users.

Bias

Platform, product-category or customer-level differences should be monitored to prevent unfair treatment.

Class Imbalance

High-risk orders represent only **8.57%** of the dataset, so accuracy alone is insufficient.

Precision, recall and F1-score should also be considered.

Data Leakage

Features unavailable at prediction time should not be used.

Human Oversight

Automated predictions should support human review rather than make irreversible decisions automatically.

29. MANAGERIAL RECOMMENDATIONS

Based on the analysis, the following recommendations are proposed:

1. **Prioritize high-risk orders** for early investigation.
2. Monitor platforms with higher concentrations of high-risk orders.
3. Investigate categories with unusual refund or complaint behaviour.
4. Track delivery delays as an important consumer-trust indicator.
5. Monitor customers with repeated refund or complaint patterns.
6. Use the Decision Tree as an interpretable benchmark.
7. Introduce real refund-processing timestamps in future data collection.
8. Add a dedicated Returns table to improve return-frequency analysis.
9. Add Seller ID and seller-level attributes for true seller-risk analysis.
10. Retrain the production model using only pre-outcome information.

30. LIMITATIONS

The project has the following limitations:

1. The supplied dataset is a single integrated CSV rather than six independent source systems.
2. Seller ID is unavailable.
3. Actual refund-processing timestamps are unavailable.
4. A separate return field is unavailable.
5. The date/time field has limitations in its original representation.
6. CTRI-derived targets introduce potential target leakage.
7. The model therefore represents an analytical benchmark rather than a production-ready predictive system.
8. Additional real-world variables such as inventory availability, payment failures and seller information would improve the framework.

31. FUTURE ENHANCEMENTS

Future work can include:

- Integrating independent Inventory data.
- Integrating Payment data.
- Adding Seller information.
- Adding Returns and Complaint tables.
- Adding actual refund timestamps.

- Implementing a leakage-free prediction target.
- Applying explainable AI techniques such as SHAP.
- Deploying real-time risk scoring.
- Adding automated alerts.
- Implementing the warehouse using PostgreSQL/MySQL.
- Connecting the warehouse directly to Power BI or another BI platform.
- Monitoring model drift over time.

32. CONCLUSION

This project developed an integrated analytical framework for consumer-trust analysis in e-commerce and quick-commerce environments.

A dataset containing **100,000 orders** was cleaned and transformed into analytical features. A **Consumer Trust Risk Index (CTRI)** was developed using complaint, refund, delivery, product, platform and customer behavioural indicators.

The analysis identified:

- **12,829 Low-risk orders**
- **78,599 Medium-risk orders**
- **8,572 High-risk orders**

A star-schema data warehouse was successfully created with one fact table and six dimension tables. All foreign-key integrity checks returned **zero invalid rows**.

Five OLAP operations—Roll-up, Drill-down, Slice, Dice and Pivot—were implemented to support multidimensional analysis.

Three machine-learning models were evaluated. The Decision Tree achieved the strongest overall benchmark performance with:

99.24% Accuracy, 92.49% Precision, 99.18% Recall, 95.72% F1-score and 99.69% ROC-AUC.

Five-fold cross-validation further produced a mean F1-score of **0.9590** for the Decision Tree.

However, because the high-risk target is derived from CTRI and some model features contribute to CTRI, the results may be optimistic due to target leakage. This limitation is explicitly recognized, and a leakage-free production implementation is recommended.